MIL-C-9912B(USAF)
19 March 1980
SUPERSEDING
MIL-C-9912A(USAF)
12 MARCH 1970

MILITARY SPECIFICATION
CUTTER, TILE, ASPHALT, PLASTIC, RUBBER

1. SCOPE

*1.1 Scope. This specification covers one type of manually operated, shear type cutter for resilient floor tile.

2. APPLICABLE DOCUMENTS

*2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bid, shall form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636 Box, Shipping, Fiberglass

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Forth Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1425) appearing at the end of this document or by letter.

FSC 5110

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STANDARDS

Federal

FED-STD-H28	Screw Thread Standards for Federal Services.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

3. REQUIREMENT

*3.1 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in

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commercial practice. None of the above shall be interpreted to mean that the use of used or rebuilt products will be allowed.

*3.2 First article. This specification includes requirements for first article testing. Unless otherwise specified (see 6.2) the supplier shall furnish one complete cutter as a preproduction sample for testing in accordance with section 4. The item may be a preproduction item or a production item which conforms to the requirements of this specification. In either case, the first article shall be manufactured in the same facilities used for manufacture of the production items. Approval of the preproduction item does not relieve the supplier of the responsibility for compliance with all provisions of this specification.

3.3 Design. The cutter shall be of the latest design currently supplied by the manufacturer. The design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

*3.3.1 Threads. All threads shall conform to requirements of FED-STD-H28.

3.3.2 Interchangeability. All parts shall be manufactured to such standards as will permit replacement of worn or damaged parts without modification of parts or the cutter.

*3.4 Construction. The cutter shall be a hand operated, compact, portable, structure with a built-in carrying handle and weighing not more than 45 pounds (20.4 kg). The cutter shall be capable of producing both square and diagonal cuts in all commercially available resilient floor tile, size 9 x 9 inch, (228 x 228 mm), and shall produce square cuts in tile size 12 x 12 inch (305 x 305 mm) up to 3/16 inch (4.8 mm) thick. Cutting blade (s) shall be hardened and ground tool steel, demountable replaceable type. The blade holder shall be equipped with wear compensating guide(s) to maintain blade alignment throughout the life of the cutter. The blade actuating hand lever shall be proportionately shaped, with a comfortable handgrip, and of sufficient length and strength to transmit the necessary cutting force to the blade. Handgrip force required to perform any cutting operation within the capacity of the cutter shall not exceed 75 pounds (34 kg).

*3.4.1 Gages. The cutter shall be equipped with tile positioning guide gages for performing both 90° ($\pi/2$ rad) (square) and 45° ($\pi/4$ rad) (angle) cuts. The gage for square cutting shall be either a base mounted fixed gage bar or a pivot type bar fitted with an index locking pin for positioning the gage in both 90° ($\pi/2$ rad) and 45° ($\pi/4$ rad) locations. In either case, the gage shall be graduated in increments of 1/8 inch (3mm) or less and marked with progressive numerical characters at each inch interval to indicate the distance from the blade cutting plane. The 45° ($\pi/4$ rad) angle gage shall include a positive type clamping device for securing the gage at any desired position within the adjustment range.

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3.4.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 Surfaces. All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.4.5 Painting The cutter shall be painted in accordance with the manufacturer's standard commercial practice.

3.5 Technical data. Unless otherwise specified (see 6.2) one copy of the supplier's instruction manual, including a detailed parts list, shall be furnished with each cutter.

3.6 Nameplate. Unless otherwise specified (see 6.2), a corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each cutter:

Nomenclature
Manufacturer's Name
Manufacturer's Model and Serial Number
Federal Stock Number
Date of Manufacture
U.S.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Inspections. Inspections of the item shall consist of first article or preproduction testing, sampling (quality conformance) inspections, and predelivery inspections. Approval of first article or preproduction testing will result in the producer becoming qualified for production of the item applicable to the purchase order requirements but it does not preclude requirements for other testing and inspections specified herein.

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*4.2.1 Preproduction or first article testing. Unless otherwise specified (see 6.2), preproduction or first article testing shall be the responsibility of the producer and shall be in accordance with paragraphs 4.2.1.1, 4.3, 4.4, 4.4.1, and 4.4.2. A failure of an item to pass requirements of preproduction or first article testing shall result in disapproval until corrective action has been taken. Requirements for preproduction or first article testing may be waived if any of the following conditions are applicable:

a. The producing manufacturer is a regular producer of item required, and the item required is for sale on a commercial basis.

b. The producing manufacturer is presently in production of the item required under a contract of which preproduction or first article testing requirements has either been waived or has been approved.

c. The producing manufacturer has previously produced the required item within 12 months preceding solicitations for this requirement.

*4.2.1.1 Test report. The supplier shall furnish a certified test report showing that the first article conforms to all requirements of this specification.

*4.2.2 Sampling for quality conformance. All cutters offered for delivery at one time shall be considered a lot for examination and test. Samples shall be taken from the inspection lot in accordance with MIL-STD-105, based on single sampling plan for normal inspection, at the level specified. Defects in excess of the acceptance number shall be cause for rejection of the representative lot. All testing and inspections, with the exception of submission of the test report, that are applicable to preproduction or first article testing are also applicable to sampling testing and inspections.

4.2.2.1 Visual and dimensional inspection. Each sample cutter shall be visually inspected for conformance with Section 3. Classification of defects shall be as specified in Table I. Sampling shall be at inspection Level I with an acceptable quality level (AQL) of 2.5 percent defective for major defects and 6.5 percent defective for total defects.

*4.2.2.2 Testing. Testing of samples shall be in accordance with Level II of MIL-STD-105.

*4.3 Inspection. Inspections applicable to the item shall be in accordance with Table I.

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TABLE I
CLASSIFICATION OF DEFECTS

Category	Defects
Major:	
101	Assembly not complete
102	Fastening devices; missing, not properly installed, not restrained
103	Actuating mechanism; excessive looseness or too tight
104	Guide gages; missing, graduation marking not as specified, ineffective clamp mechanism
105	cutting blade; not aligned with shear plate, loose in holder, dull
106	Casting defects; pits, modules, and sand cavities
107	Defective welding
Minor:	
201	Finish; burrs, sharp edges, scale or flux
202	Painting; chipped, flaked, peeling
203	Marking; missing, incomplete, illegible

4.4 Tests.

*4.4.1 Operational tests. Operate the blade control lever on cutter, to actuate the blade at least twelve cycles, from the extreme open to the extreme closed positions. The blade travel shall range from not less than 1/4 inch (6 mm) above the shear plate in the open position to not less than 1/16 inch (1.6 mm) below the shear plate in the closed position. Evidence of missing parts, faulty assembly excessive tightness or blade travel range less than specified shall be cause of rejection. For sampling inspections, the QAL shall be 2.5 percent defective.

*4.4.2 Performance test. The cutter shall perform at least fifteen (15) distinct cutting operations. The number of separate test pieces, type of material and thickness variation required for this test shall be as follows: 4 each 9 x 9 inch (228 x 228 mm) vinyl 3/32 inch (2.4 mm) thick, 4 each 9 x 9 inch (228 x 228 mm) asphalt 1/8 inch (3 mm) thick, 4 each 9 x 9 inch (228 x 228 mm) vinyl 3/16 inch (4.8 mm) thick, and 1 each 12 x 12 inch (305 x 305 mm) vinyl 1/8 inch (3 mm) thick. Two pieces of each type and thickness of the 9 x 9 inch (228 x 228 mm) size shall be cut at both 45° ($\pi/4$ rad) and 90° ($\pi/2$ rad) positions and three square cuts

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shall be performed on the 12 x 12 inch (305 x 305 mm) size. The cut edge shall be smooth, straight, free from splits, cracks, and broken edges and shall not deviate from a true plane in relation to the applicable guide gage by more than 1/32 inch (0.8 mm) in twelve inches (305 mm). For sampling inspections, the AQL shall be 1.5 percent defective.

*4.5 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of Section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

5. PACKAGING

*5.1 Preservation and packaging. Packaging shall be Level A or commercial as specified (see 6.2).

*5.1.1 Level A. The cutter shall be cleaned and coated with a light or thin film preservative. The cutter shall then be covered with a greaseproof wrap.

*5.1.2 Commercial. The cutter shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the cutter against damage during shipment, handling and storage.

*5.2 Packing. Packing shall be Level A or commercial as specified (see 6.2).

*5.2.1 Level A. The cutters, packaged as specified in 5.1.1, shall be packed in snug-fitting fiberboard boxes conforming to PPP-B-636, Class weather resistant. The boxes shall be closed, waterproofed, and strapped or reinforced in accordance with the appendix of the box specification.

*5.2.2 Commercial. The cutters, packaged in accordance with 5.1.2, shall be packed in fiberboard boxes to insure delivery at destination, to provide for redistribution by the initial receiving activity and shall be acceptable by common carrier under the National Motor Freight Classification and the Uniform Freight Classification.

*5.3 Marking. Marking shall be in accordance with the contract or order.

6. NOTES

*6.1 Intended use. The cutter covered by the specification is intended for cutting all types of commercially available resislient floor tile in thickness range from 1/16 inch (1.6 mm) to 3/16 inch (4.8 mm).

*6.2 Ordering data. Procurement documents shall specify the following:

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a. Title, number and date of this specification along with its latest applicable amendment.

b. Identification of item required (see 6.3).

c. When first article is not required (see 3.2).

d. Technical data if different (see 3.5).

e. Nameplate if different (see 3.6).

f. Levels of packaging and packing required (see 5.1 and 5.2).

6.3 Item identification. The identtification of the item required by use of this specification will be M9912 (see 6.2).

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire contents irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-9912B (USAF), Cutter, Tile, Asphalt, Plastic, Rubber

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